Causal Discovery and Average Treatment Effect Estimation on NHANES

Technical Project Report for PhD in Mechanism-Informed Multimodal AI u^b

Ömer Alkaya

July 2026

Repository: https://github.com/oalkaya/Causal-NHANES

Contents

1 Relevant theoretical background

The course material was used to identify the theoretical requirements most relevant to the implementation. The first seven lectures of Brady Neal's *Introduction to Causal Inference* provided the potential-outcomes and graphical foundations, while the observational causal-discovery lecture covered PC and functional causal models.[1]

2 NHANES data preprocessing

2.1 Analytic variables

The raw input contains 3,842 observations over fourteen variables such as BMI, demographic context, activity and diet, anthropometry, and blood pressure. The cleaned dataset is numeric and contains no missing or infinite values before causal discovery.

Table 1: NHANES variables categorized into groups.

Group	Variables
Demographics	Age, Gender, Income Ratio
Activity and diet	Vigorous Activity, Sedentary Minutes, Total Calories, Protein, Carbohydrates, Total Sugars, Dietary Fiber
Anthropometry	BMI, Waist Circumference
Blood pressure	Systolic BP, Diastolic BP

2.2 NHANES sanity checks and exclusions

The range rules are conservative sanity filters rather than clinical diagnostic thresholds. Their purpose is to detect impossible or strongly implausible data.

Table 2: Preprocessing validity rules and observed exclusions.

Variable	Accepted values or interval	Invalid rows
Age	$0 \leq x \leq 120$	0
Gender	$x \in \{1, 2\}$	0
Income Ratio	$0 \leq x \leq 10$	0
BMI	$10 \leq x \leq 80$	0
Waist Circumference	$40 \leq x \leq 200$	0
Vigorous Activity	$x \in \{0, 1\}$	0
Sedentary Minutes	$0 \leq x \leq 1440$	15
Systolic BP	$50 \leq x \leq 300$	0
Diastolic BP	$30 \leq x \leq 200$	27
Total Calories	$0 \leq x \leq 15000$	0
Protein (g)	$0 \leq x \leq 600$	0
Carbohydrates (g)	$0 \leq x \leq 1500$	0

[1] Course page: `https://www.bradyneal.com/causal-inference-course`.

Variable	Accepted values or interval	Invalid rows
Total Sugars (g)	$0 \leq x \leq 1000$	0
Dietary Fiber (g)	$0 \leq x \leq 200$	0

No selected variable became missing after numeric conversion. Fifteen rows failed the sedentary-minutes range by being set to 9999 and 27 rows failed the diastolic-pressure range by being set to $\tilde{0}$. In total, 42 rows were removed, leaving 3,800 observations.

Table 3: Preprocessing outcome.

Measure	Count
Original observations	3,842
Clean observations	3,800
Dropped observations	42

The cleaned variables are standardized inside the causal-discovery scripts because PC's Fisher-z tests and regression-based discovery are easier to compare numerically on a common scale. The inference script does *not* standardize the outcome. Instead, it rescales only the continuous treatments into reporting units so that coefficients are directly interpretable as effects per 100 kcal or per 10 g.

3 Causal discovery and graph selection

3.1 Choice of discovery methods

The three discovery methods were selected to represent the main methodological settings covered in Brady Neal's causal-discovery lectures: independence-based discovery, linear models with non-Gaussian disturbances, and nonlinear additive-noise models. **PC** was chosen as the constraint-based method, **DirectLiNGAM** as the linear non-Gaussian method, and **RESIT** as the nonlinear additive-noise method. This provides one representative algorithm from each family and allows methods based on substantially different identifying assumptions to be compared.

3.2 Selection criteria

The goal of discovery was not simply to obtain a graph with a high number of statistically supported edges. A graph used for the next stage had to satisfy four practical requirements:

1. **Acyclicity:** downstream backdoor identification assumes a DAG.
2. **Temporal and domain plausibility:** baseline or earlier-domain variables should not be caused by later cardiometabolic measurements.
3. **Treatment connectivity:** each of the four predefined treatments must have a directed path to BMI; otherwise the graph encodes no structural route for the corresponding total-effect query.
4. **Sparseness:** among graphs satisfying the first three requirements, a less dense graph is preferred because it introduces fewer opportunities for spurious paths and unnecessarily large adjustment sets.

3.3 Initial unconstrained PC experiment

PC first learns an undirected skeleton from conditional-independence relations and then orients edges using identified collider structures and subsequent orientation rules. Because the cross-sectional data do not determine every direction unambiguously, the unconstrained results sometimes placed BMI or blood-pressure variables upstream of dietary, behavioral, or demographic variables. These orientations were undesirable for the upcoming ATE estimations.

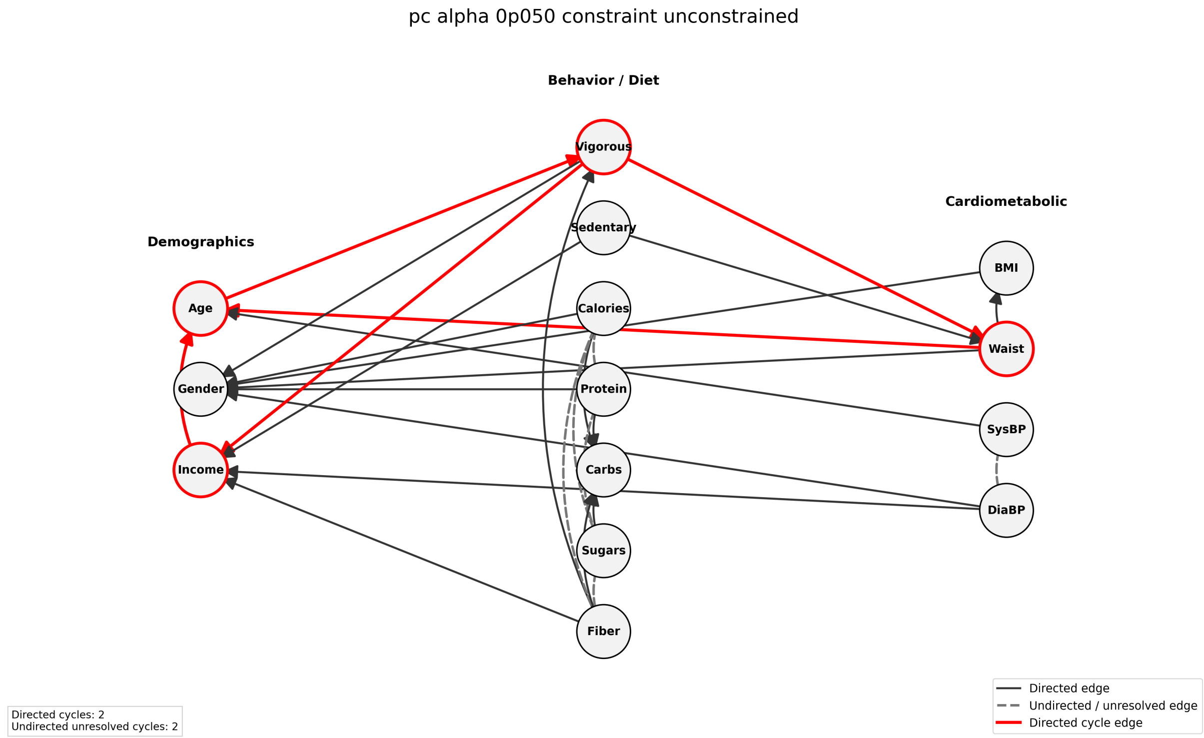

Figure 1: Unconstrained PC at $\alpha = 0.05$. The exported graph contains directed cycles and undesirable connections into the demographic variables. Unresolved edges are also present, illustrating that observational conditional-independence information does not generally identify one fully oriented DAG.

3.4 Moderate constraints

To prevent undesirable orientations observed in the unconstrained PC results from carrying into the downstream causal-inference stage, an initial set of background-knowledge constraints was introduced. The variables were assigned to three broad tiers representing the assumed ordering.

- **Tier 0 – Demographic:** Age, Gender, and Income Ratio.
- **Tier 1 – Behavioral and Dietary:** Vigorous Activity, Sedentary Minutes, Total Calories, Protein, Carbohydrates, Total Sugars, and Dietary Fiber.
- **Tier 2 – Cardiometabolic:** BMI, Waist Circumference, Systolic Blood Pressure, and Diastolic Blood Pressure.

Edges from a later tier into an earlier tier were forbidden. For example, body measurements and blood-pressure variables could not be oriented as causes of demographic, dietary, or activity variables. Edges within the same tier remained allowed, and no individual edge was required to appear.

This configuration was intentionally moderate. It imposed only the broad ordering needed to obtain graphs compatible with the planned treatment-to-BMI queries, while leaving the discovery algorithms free to determine relationships among variables within the same domain.

Table 4: Discovery methods and how constraints enter the implementation.

Method	Main modeling assumption	Constraint implementation
PC	Conditional-independence structure, causal Markov and faithfulness assumptions; Fisher-z test in the implementation	Forbidden and required *direct* relations are supplied to causal-learn's background-knowledge object during search and orientation. The output may remain partially oriented.
DirectLiNGAM	Linear acyclic structural equations with non-Gaussian disturbances and causal sufficiency	Forbidden directions are converted to a prior-knowledge matrix using `no_paths`. This is stronger than forbidding one direct edge: it prevents the source from being an ancestor of the target. Learned coefficients below a threshold are omitted, but remaining edges are not reoriented post hoc.
RESIT	Nonlinear additive-noise model; regression followed by residual-independence testing	The same prior-knowledge matrix and ancestral `no_paths` semantics are supplied natively. Random forests are used for regression. No learned edge is added, deleted, or reoriented after fitting merely to match the desired graph.

As summarized in Table 4, the three discovery methods differ in how they incorporate background knowledge. Only PC supports required direct-edge constraints in the implementation used here. DirectLiNGAM and RESIT therefore rely on forbidden-direction constraints only, and the required-edge entries in the YAML configurations remain inactive for these experiments.

3.5 Moderate-constraint results

The moderate constraint configuration was evaluated across multiple significance levels and coefficient thresholds for all three discovery methods. The purpose of this stage was to determine whether the broad tier ordering alone could produce a candidate DAG that both supported all four treatment-to-BMI queries and remained sufficiently sparse for interpretation and downstream adjustment-set identification.

Table 5 presents a small set of representative candidates that illustrate the main trade-offs observed during the search. The complete collection of generated graphs, edge lists, and summary files is available in the project repository.

Table 5: Representative candidates from the moderate-constraint experiments. "All paths" indicates that all four treatment variables have a directed path to BMI.

Method	Hyperparameter	Edges	All paths	Main observation
PC	$\alpha = 0.05$	26	**No**	None of the four treatment variables has a directed path to BMI.
PC	$\alpha = 0.70$	38	**No**	Vigorous Activity is connected to BMI, but the three dietary treatments are not all connected.
DirectLiNGAM	threshold 0.060	41	**Yes**	All four queries are supported, but the graph remains comparatively dense.
DirectLiNGAM	threshold 0.070	35	**No**	The graph is sparser, but the directed path from Total Calories to BMI disappears.
RESIT	$\alpha = 0.001$	84	**Yes**	All four queries are supported, but the resulting DAG is close to saturated.

The three methods exhibited different limitations. PC produced the sparsest candidates, but the resulting orientations did not provide directed paths from all four treatments to BMI. DirectLiNGAM came closest to the desired balance: at a threshold of 0.060, all four treatment paths were present, whereas increasing the threshold to 0.070 reduced the edge count but removed the Total Calories path. RESIT consistently retained the required treatment connectivity, but produced substantially denser graphs.

With fourteen variables, a DAG can contain at most

$$\frac{14(14-1)}{2} = 91$$

directed edges. The representative RESIT graph contains 84 edges and is therefore close to the maximum possible density. The concern is not the edge count alone. A nearly saturated DAG encodes many possible causal and backdoor paths, is difficult to interpret, and can lead to more complicated adjustment requirements. Conditioning on additional variables that are not necessary for identification may also weaken positivity and overlap across treatment levels, increase reliance on extrapolation, and reduce statistical precision.

Figures 2–4 show one representative graph with moderate constraints from each discovery method.

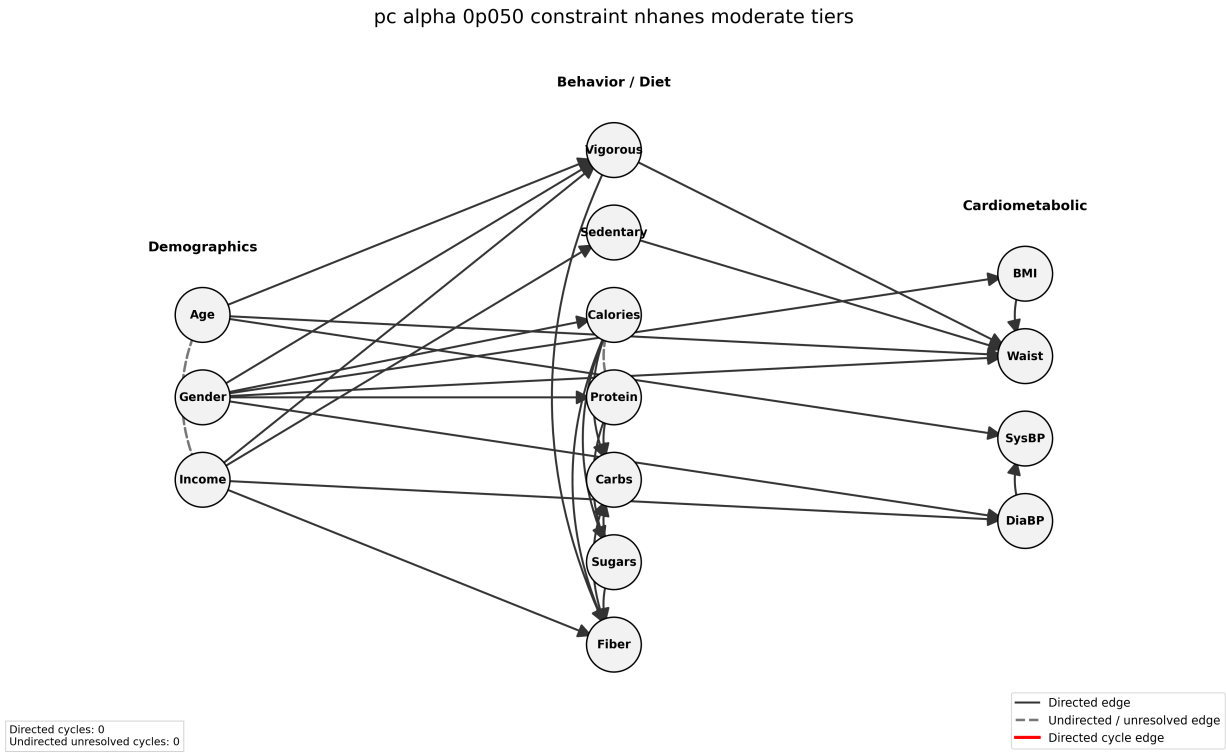

Figure 2: Moderate PC, $\alpha = 0.05$: 26 edges, with no directed path from the four target treatments to BMI.

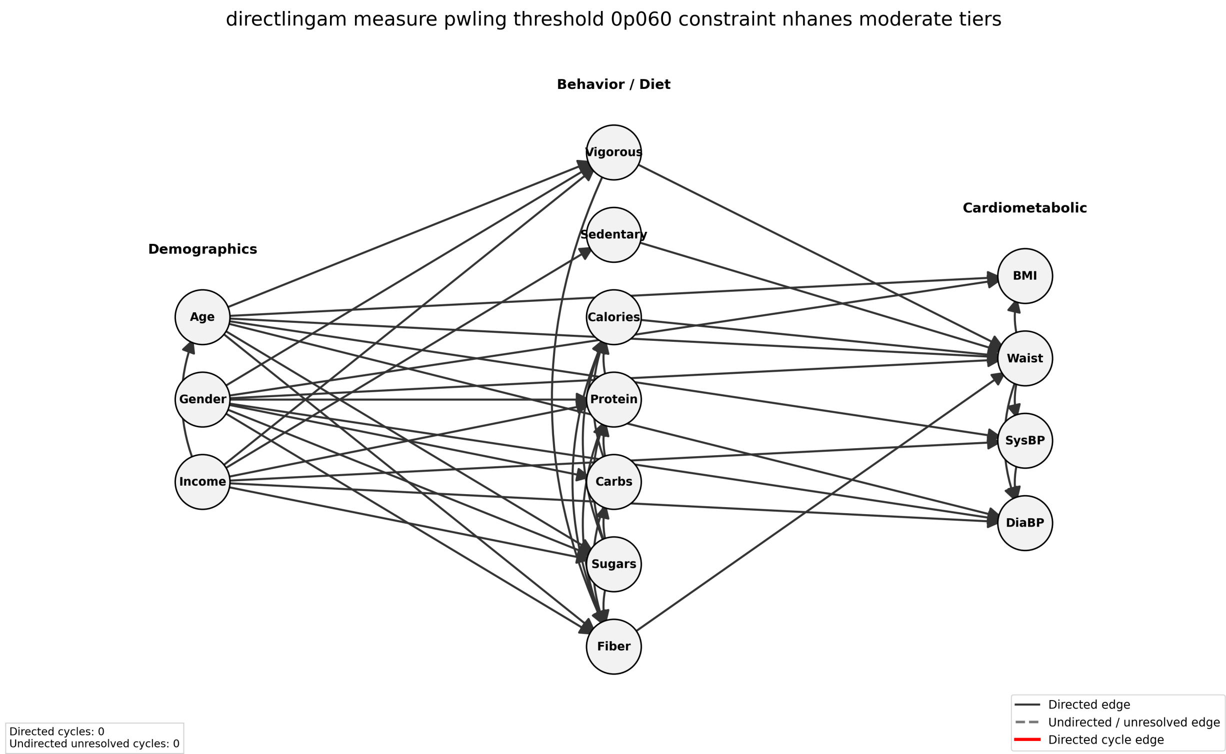

Figure 3: Moderate DirectLiNGAM, threshold 0.060: 41 edges and directed paths from all four treatments to BMI.

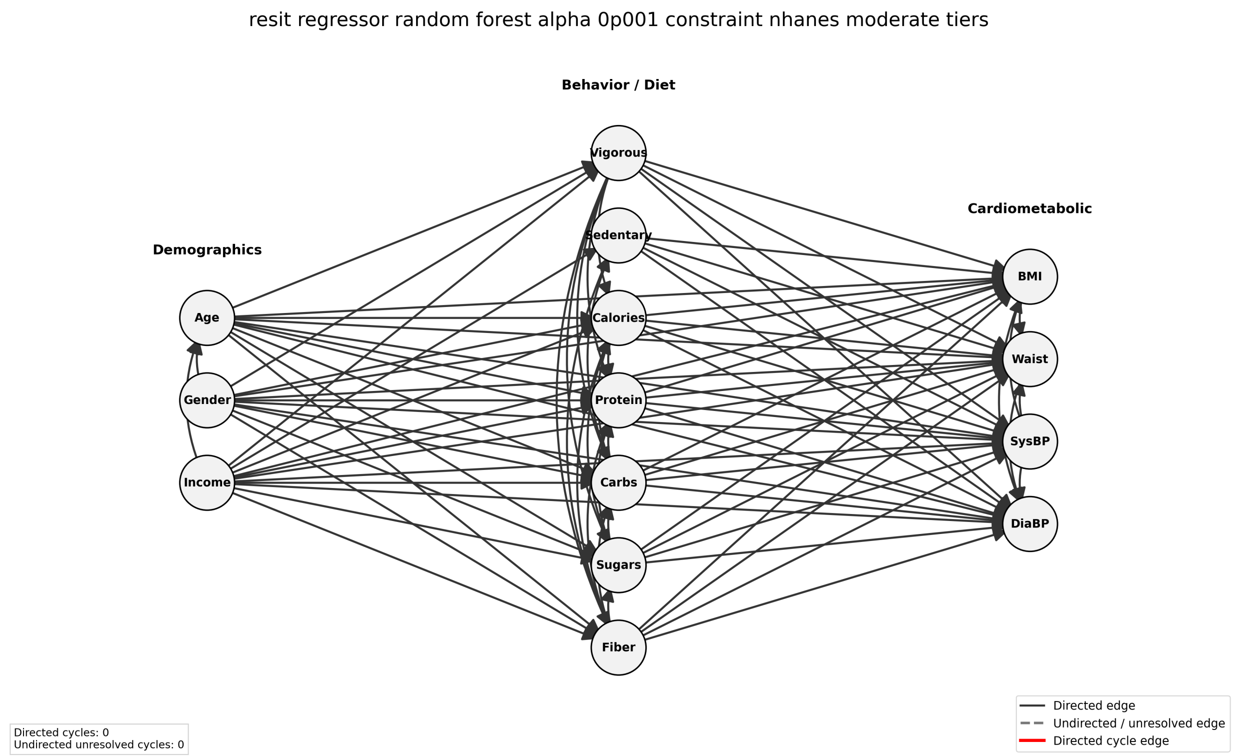

Figure 4: Moderate RESIT, $\alpha = 0.001$: 84 edges and directed paths from all four treatments to BMI.

3.6 Strict constraints

The strict configuration refines the earlier three-tier ordering into five more specific levels:

Table 6: Variable ordering used in the strict constraint configuration.

Tier	Group	Variables
0	Demographics	Age, Gender
1	Socioeconomic status	Income Ratio
2	Behavior and diet	Vigorous Activity, Sedentary Minutes, Total Calories, Protein, Carbohydrates, Total Sugars, Dietary Fiber
3	Body measurements	Waist Circumference, BMI
4	Blood pressure	Systolic Blood Pressure, Diastolic Blood Pressure

In addition to the tier ordering, the strict configuration forbids the direct edges BMI $\rightarrow$ Waist Circumference, Age $\rightarrow$ BMI, Gender $\rightarrow$ BMI, Age $\rightarrow$ Waist Circumference, and Gender $\rightarrow$ Waist Circumference. These additional restrictions were chosen to favor candidate graphs compatible with the next stage of the analysis, in which the selected dietary and activity treatments lie upstream of BMI and retain directed paths to it.

Table 7: Representative strict-constraint candidates

Method	Hyperparameter	Edges	All paths	Assessment
PC	$\alpha = 0.01$	24	No	Three treatment paths exist, but Carbohydrates has no directed route to BMI.
PC	$\alpha = 0.70$	37	No	Vigorous Activity, Protein and Carbohydrates are connected; Total Calories is not.
RESIT	$\alpha = 0.001$	84	Yes	Connected but near-saturated.
DirectLiNGAM	threshold 0.070	33	Yes	Acyclic, all four paths present, and substantially less dense; **selected**.
DirectLiNGAM	threshold 0.080	27	No	Sparser, but only the Vigorous Activity path remains.

The strict DirectLiNGAM graph at threshold 0.070 is the first candidate along the increasing-threshold path that retains all four treatment-to-BMI routes while reducing the graph to 33 edges. Its density is $33/91 \approx 36.3\%$, compared with $84/91 \approx 92.3\%$ for the least dense all-path RESIT example shown above.

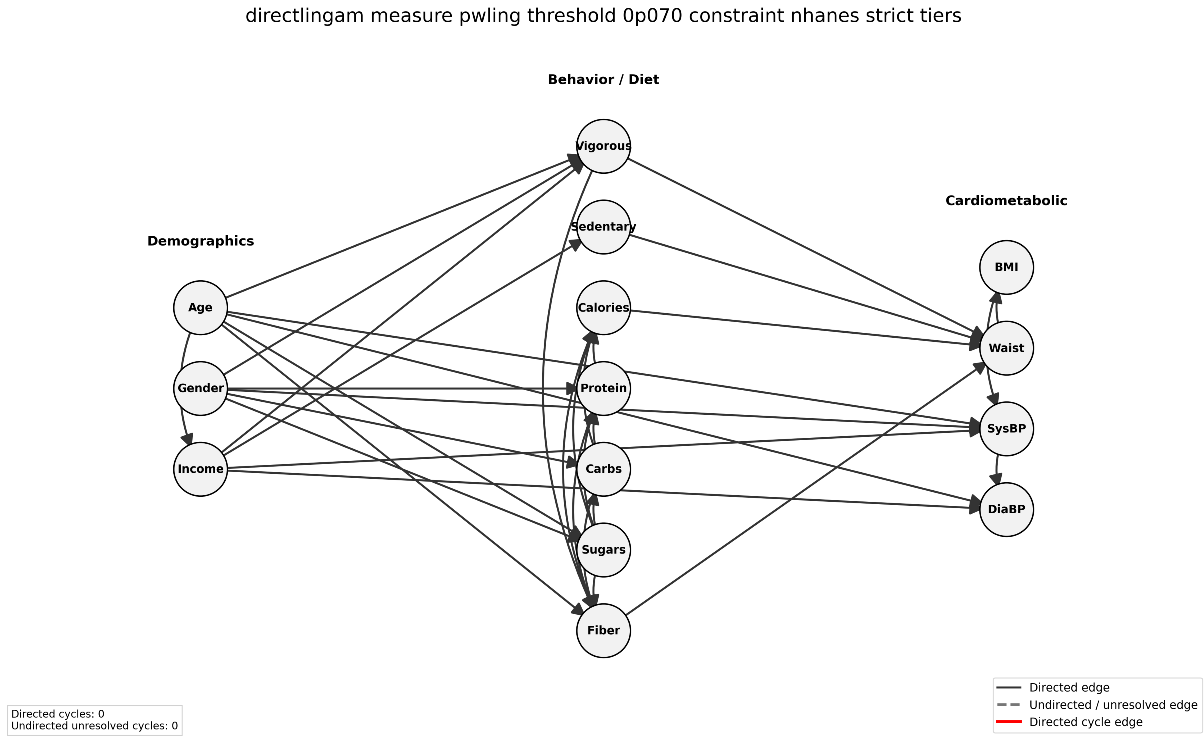

Figure 5: Selected strict DirectLiNGAM graph at coefficient threshold 0.070. All four treatment variables have directed paths to BMI, the graph is acyclic, and the edge count is 33.

The selected CSV is

`outputs/directlingam/graph_csvs/directlingam_measure_pwling_threshold_0p070_constraint_nhanes_strict_tiers_edges.csv`.

4 Adjustment sets and causal effect estimation

4.1 Target estimands and reporting units

For each treatment, the target is the population-average total effect on BMI for the contrast listed below.

Table 8: Treatment contrasts used in estimation.

Treatment	Type	Reported contrast
Vigorous Activity	Binary	$0 \to 1$
Total Calories	Continuous	Increase of 100 kcal
Protein	Continuous	Increase of 10 g
Carbohydrates	Continuous	Increase of 10 g

Continuous treatments are rescaled before modeling so that one model unit corresponds to 100 kcal or 10 g.

4.2 Custom minimal adjustment-set finder

A custom utility was implemented to make adjustment selection explicit and independently testable. For a treatment T, outcome Y, and DAG G, the procedure is:

1. construct the treatment-specific backdoor graph by removing all edges leaving the treatment;
2. define eligible adjustment variables as ancestors of the treatment or outcome in the original DAG, excluding the treatment, outcome, and all descendants of the treatment;
3. enumerate candidate subsets in increasing order of size;
4. retain candidates that d-separate the treatment and outcome in the backdoor graph;
5. discard supersets of previously identified valid sets, yielding all inclusion-minimal backdoor adjustment sets.

4.3 Minimal sets and manual selection

Table 9: Chosen manual sets and DoWhy graph-derived sets. Every manual set was validated against the selected DAG before estimation.

Treatment	Manually selected minimal set	DoWhy-selected set
Vigorous Activity	Age, Gender, Income Ratio	Age, Gender, Sedentary Minutes
Total Calories	Dietary Fiber, Income Ratio, Vigorous Activity	Dietary Fiber, Sedentary Minutes, Vigorous Activity
Protein	Carbohydrates, Gender, Total Sugars	Carbohydrates, Gender, Total Sugars
Carbohydrates	Gender, Total Sugars	Gender, Total Sugars

The graph admits multiple inclusion-minimal sets for Vigorous Activity and Total Calories. The chosen manual sets use baseline socioeconomic variables where possible and provide an interpretable alternative to the automatic sets. This is a choice among sets that are graphically valid relative to the selected DAG; it does not establish that the selected covariates are the uniquely correct substantive confounders.

As a sanity check, the manual adjustment set implementation rejects a set if it contains an unknown variable, the treatment, the outcome, or a treatment descendant, or if it fails the d-separation test in the treatment-specific backdoor graph.

4.4 DoWhy estimation model

DoWhy is used to separate causal identification from statistical estimation. In automatic mode, the selected DAG is supplied to `CausalModel`, and `identify_effect` obtains a backdoor estimand. In manual mode, the graph-validated set is supplied as the common-cause set. Both modes use

```
method_name="backdoor.linear_regression",    target_units="ate".
```

The estimator fits a linear outcome model containing the treatment and selected adjustment variables.

4.5 Disabling unintended conditional-effect estimation

The initial setup produced conditional treatment effects through automatically generated interaction terms. Passing an empty effect-modifier list to the DoWhy model disables this behavior, so the final implementation estimates one marginal ATE per treatment.

4.6 Manual and automatic adjustment set ATE estimates

Table 10: ATE estimates from the selected DAG. Confidence intervals are 95%.

Treatment	Set mode	ATE	CI low	CI high	p-value
Vigorous Activity	Manual	-1.6960	-2.2518	-1.1402	2.402×10^{-9}
	Automatic	-1.7779	-2.3197	-1.2362	1.393×10^{-10}
Total Calories	Manual	0.0316	0.0057	0.0575	1.664×10^{-2}
	Automatic	0.0266	0.0009	0.0523	4.250×10^{-2}
Protein	Manual	0.0446	-0.0194	0.1086	1.722×10^{-1}
	Automatic	0.0446	-0.0194	0.1086	1.722×10^{-1}
Carbohydrates	Manual	-0.0349	-0.0682	-0.0016	3.987×10^{-2}
	Automatic	-0.0349	-0.0682	-0.0016	3.987×10^{-2}

Under the selected graph and linear adjustment model, Vigorous Activity is estimated at about 1.70 lower BMI points. The estimate per 100 kcal is approximately 0.032 BMI units; the Protein interval includes zero; and the Carbohydrates estimate is approximately -0.035 BMI units per 10 g. These are graph- and model-conditional estimates rather than experimental or clinical conclusions.

Manual and automatic sets produce the same Protein and Carbohydrates estimates and similar results for Vigorous Activity and Total Calories. This provides a limited specification check, but does not address unmeasured confounding or graph misspecification. Optional DoWhy refuters test selected instabilities through random-common-cause, placebo-treatment, and data-subset checks.

A Implementation map

Table 11: Repository components used in the report.

Component	Role
`utils/preprocess_nhanes.py`	Numeric conversion, plausibility checks, row-level exclusion reasons, clean data and report generation.
`scripts/run_pc.py`	Unconstrained and constrained PC experiments over significance levels.
`scripts/run_directlingam.py`	PW-LiNG DirectLiNGAM experiments over coefficient thresholds with native prior knowledge.
`scripts/run_resit.py`	Random-forest RESIT experiments with residual-independence testing and prior knowledge.
`utils/constraints.py`	Shared YAML parsing and construction of allowed, forbidden and required edge definitions.
`utils/check_treatment_paths.py`	DAG check and directed-path check from each treatment to BMI.
`utils/visualize_graph_csv.py`	Graph rendering and cycle reporting.
`utils/find_adjustment_sets.py`	Treatment-specific backdoor graph, d-separation testing and inclusion-minimal set search.
`utils/test_adjustment_sets.py`	Sanity tests for confounders, colliders, mediators and alternative blockers.
`scripts/run_inference.py`	Treatment scaling, graph/manual-set validation, DoWhy identification, ATE estimation and optional refuters.

B References

References

[1] Brady Neal. *Introduction to Causal Inference from a Machine Learning Perspective*. Course and book draft, 2020. https://www.bradyneal.com/causal-inference-course.

[2] Peter Spirtes and Clark Glymour. An algorithm for fast recovery of sparse causal graphs. *Social Science Computer Review*, 9(1):62–72, 1991. https://doi.org/10.1177/089443939100900106.

[3] Shohei Shimizu, Takanori Inazumi, Yasuhiro Sogawa, Aapo Hyvärinen, Yoshinobu Kawahara, Takashi Washio, Patrik O. Hoyer, and Kenneth Bollen. DirectLiNGAM: A direct method for learning a linear non-Gaussian structural equation model. *Journal of Machine Learning Research*, 12(33):1225–1248, 2011. https://www.jmlr.org/papers/v12/shimizu11a.html.

[4] Jonas Peters, Joris M. Mooij, Dominik Janzing, and Bernhard Schölkopf. Causal discovery with continuous additive noise models. *Journal of Machine Learning Research*, 15(58):2009–2053, 2014. https://www.jmlr.org/papers/v15/peters14a.html.